

2(a)	change in displacement / time (taken)	B1
2(b)(i)	horizontal velocity = $22 \times \cos 40^\circ$	C1
	time taken = $36 / (22 \times \cos 40^\circ)$ = 2.1 s	A1
2(b)(ii)	$u = 22 \times \sin 40^\circ$ = 14 m s^{-1}	A1
2(b)(iii)	$s = ut + \frac{1}{2}at^2$ = $(14 \times 2.1) + (\frac{1}{2} \times -9.81 \times 2.1^2)$	C1
	= 7.8 (m)	C1
	(therefore) height of wall = $7.8 + 1.2$ = 9.0 m	A1

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Question	Answer	Marks
2(b)(iii)	or	
	<i>other methods possible e.g.</i> time to maximum height = $(0 - 14) / -9.81$ (= 1.43 s) time from maximum height to wall = $2.1 - 1.43$ (= 0.67 s) maximum height above release = $(14 \times 1.43) + (\frac{1}{2} \times -9.81 \times 1.43^2)$ = 9.99 (m)	(C1)
	height from maximum to wall = $0.5 \times 9.81 \times 0.67^2$ (= 2.2 m) height above release = $9.99 - 2.2$ = 7.8 (m)	(C1)
	height of wall = $1.2 + 7.8$ = 9.0 m	(A1)